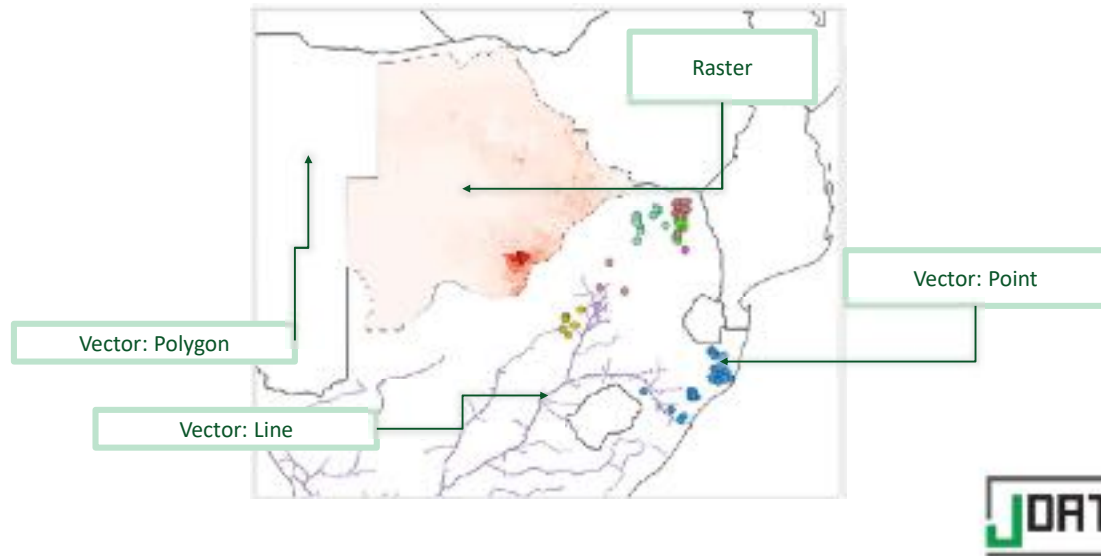


Data Types in QGIS



SECTION 3

- Basic qGIS operations



Basic operations

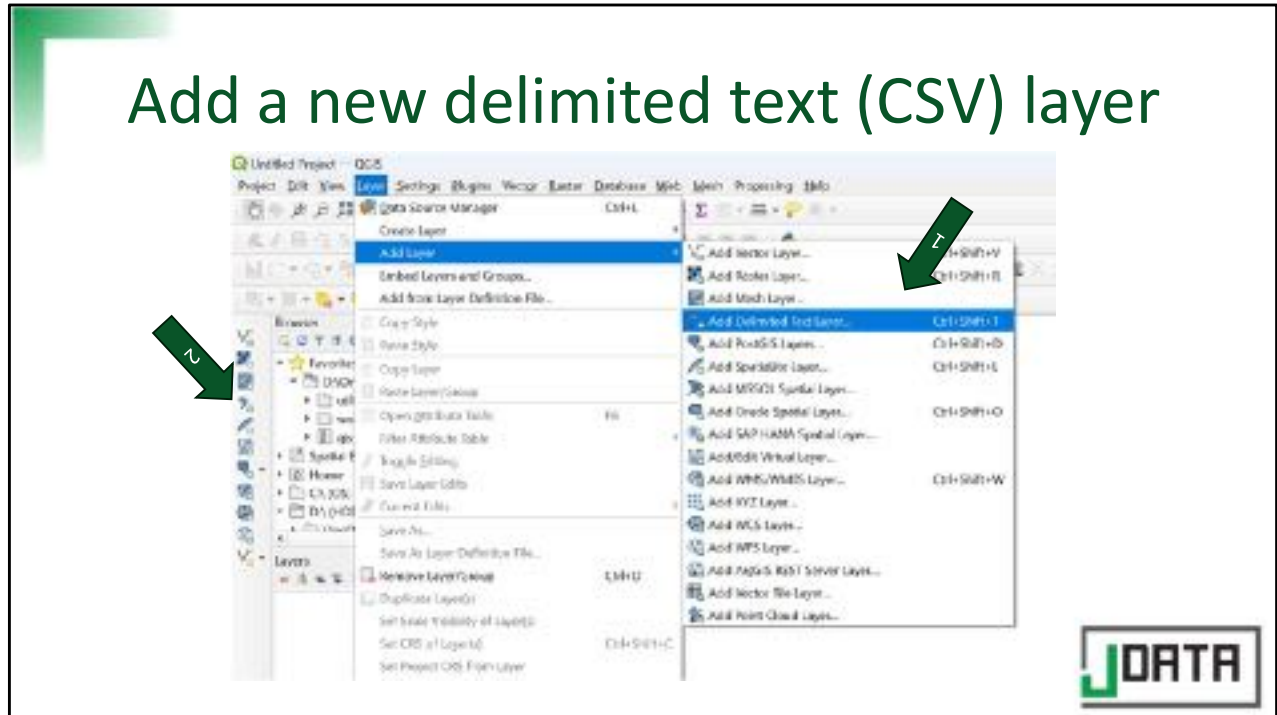
3.1: Adding vector layers

3.2: Symbolising layers

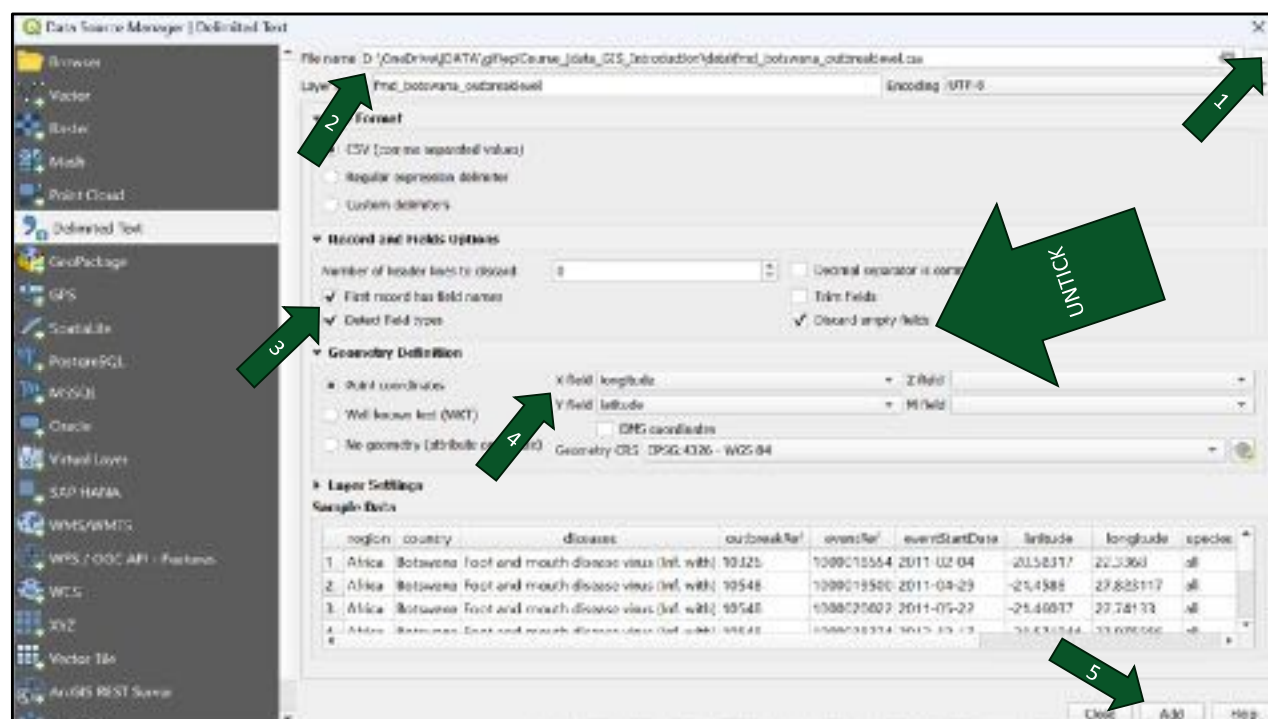
3.3 Adding labels

3.4 Vector classification

Add a new delimited text (CSV) layer



- Either add a new layer through the menu (option 1) or click the comma on the left (option 2)

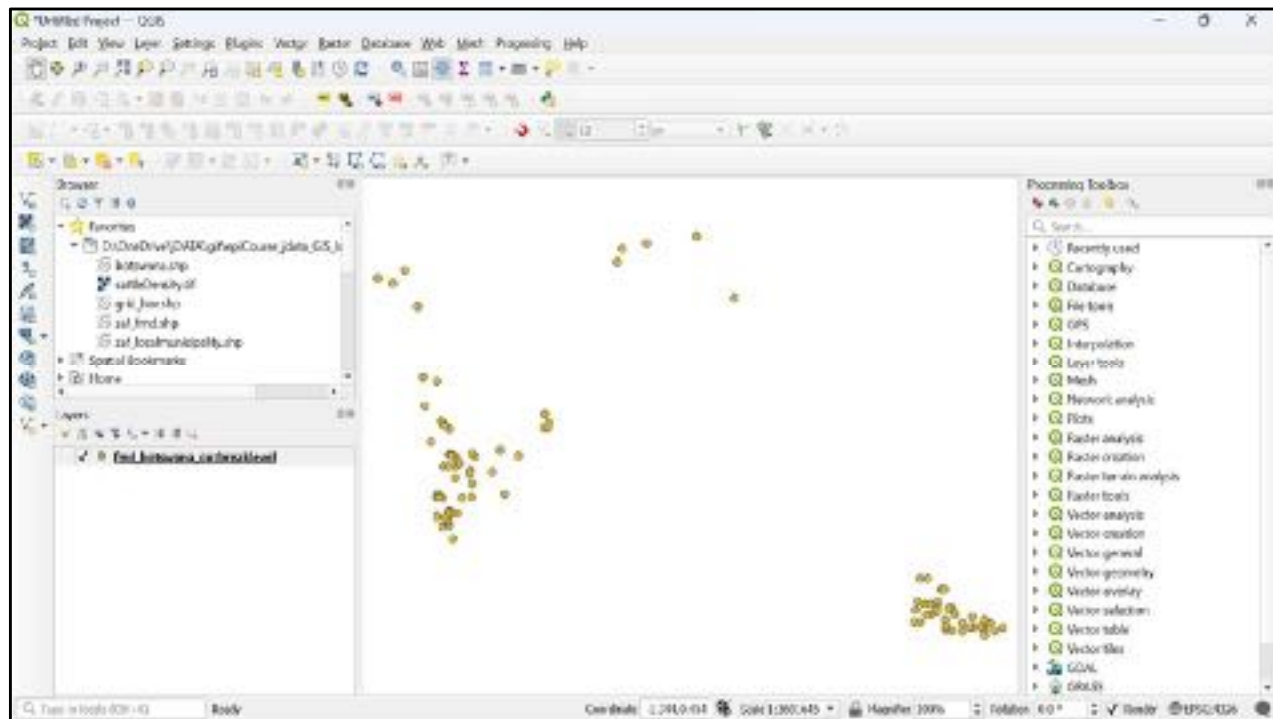


- 1 - Browse for the CSV file you wish to import
- Ensure file format is CSV
- 2 – Check the layer name and change if you wish
- 3 - If your CSV has headers then leave this ticked; Also tick the *DETECT FIELD TYPES*
- 4. Insert the longitude (X) and latitude (Y) coordinates and leave the Geometry CRS generally as WGS 84 (5)
- Click Add and Close

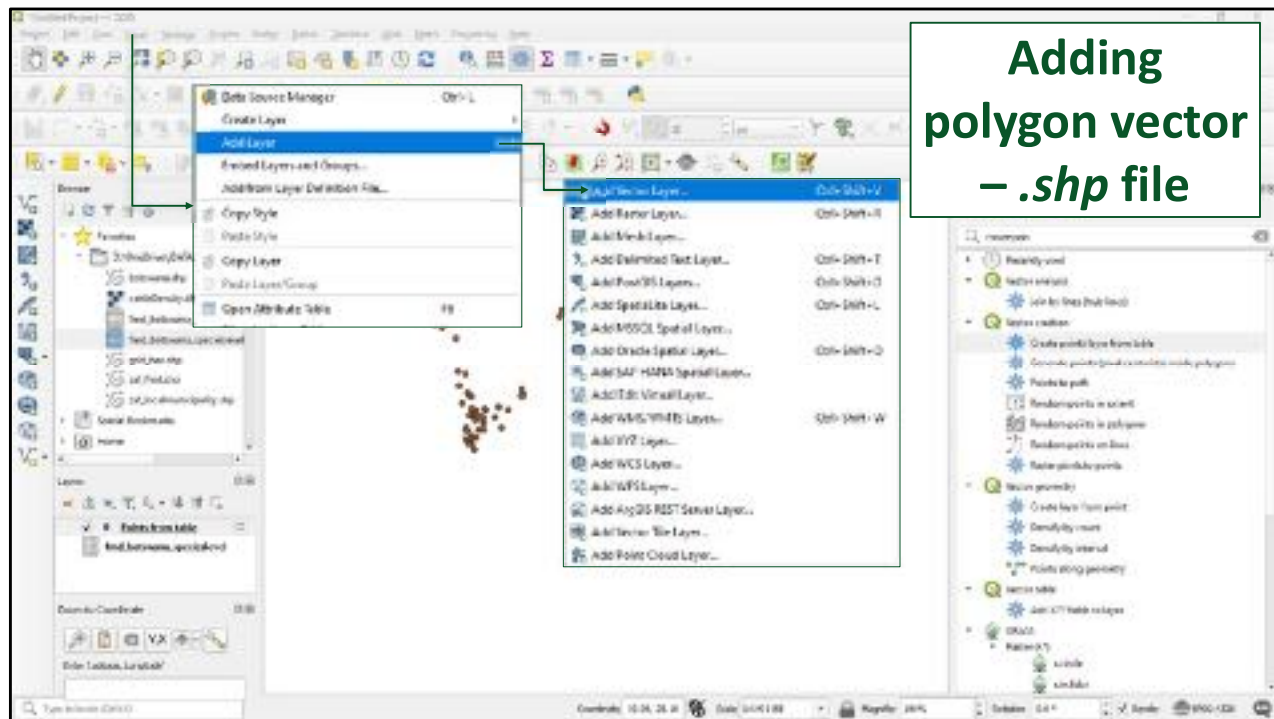
If your CSV is weird 😊

- Ensure your system settings are correct
 - Hit WINDOWS KEY
 - Type Region and select REGION
 - Look for additional setting
 - Set your decimal separator to . (that's a dot/full stop)
 - Set your list separator to **comma**

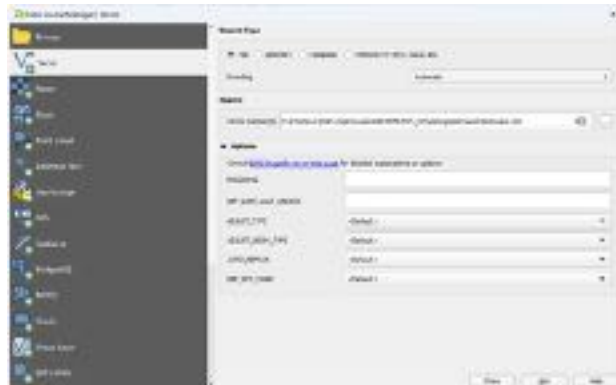




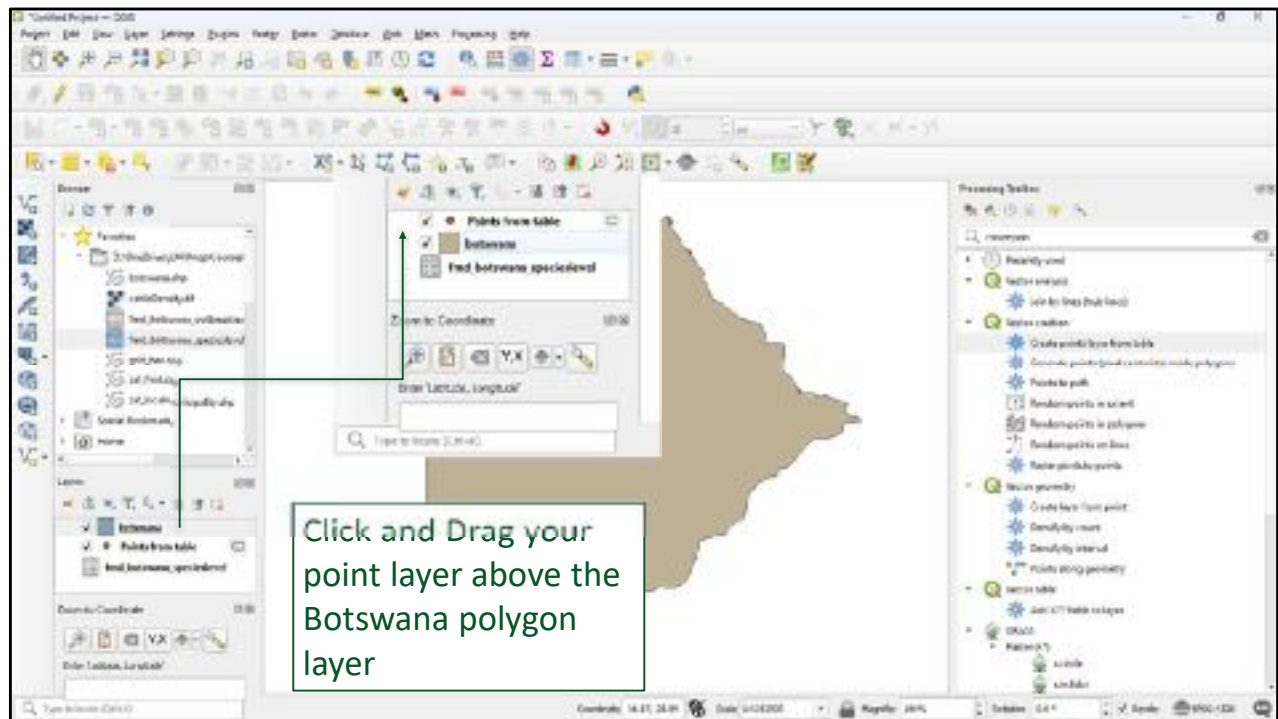
- Your map should look something like this



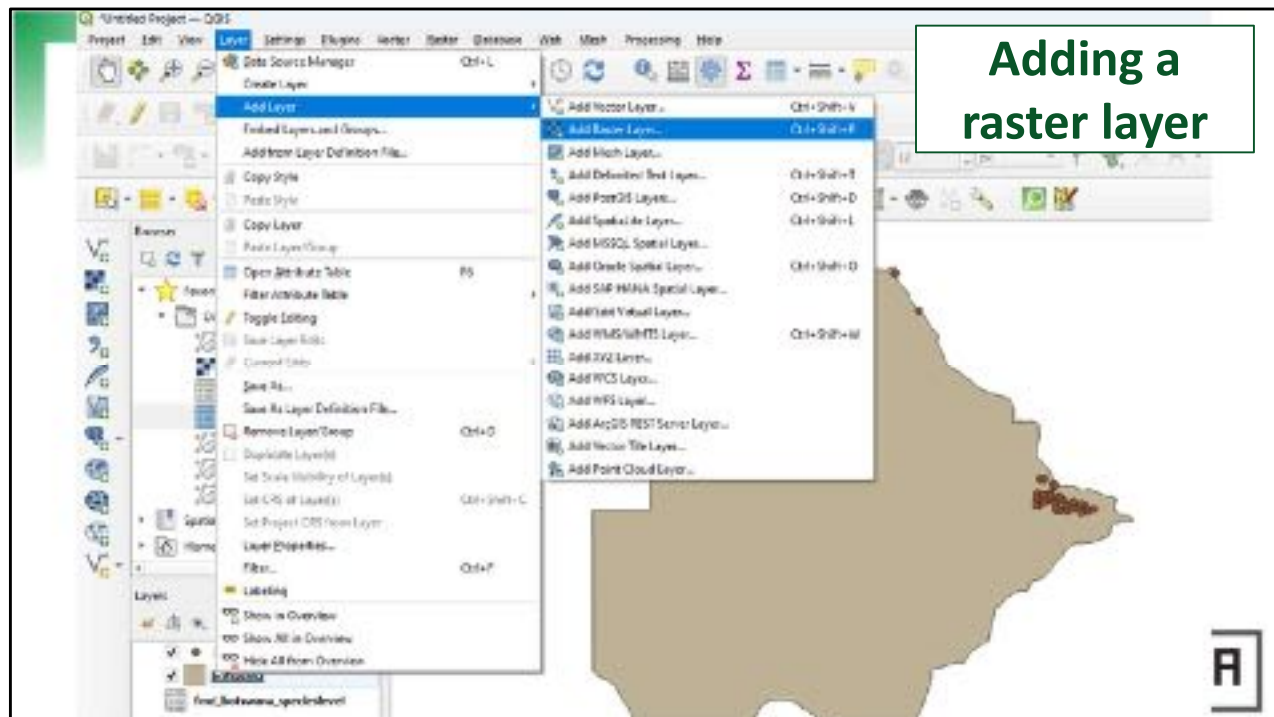
Adding polygon vector – .shp file



- Select the .shp file from the project folder
- Leave defaults and hit Add and then Close

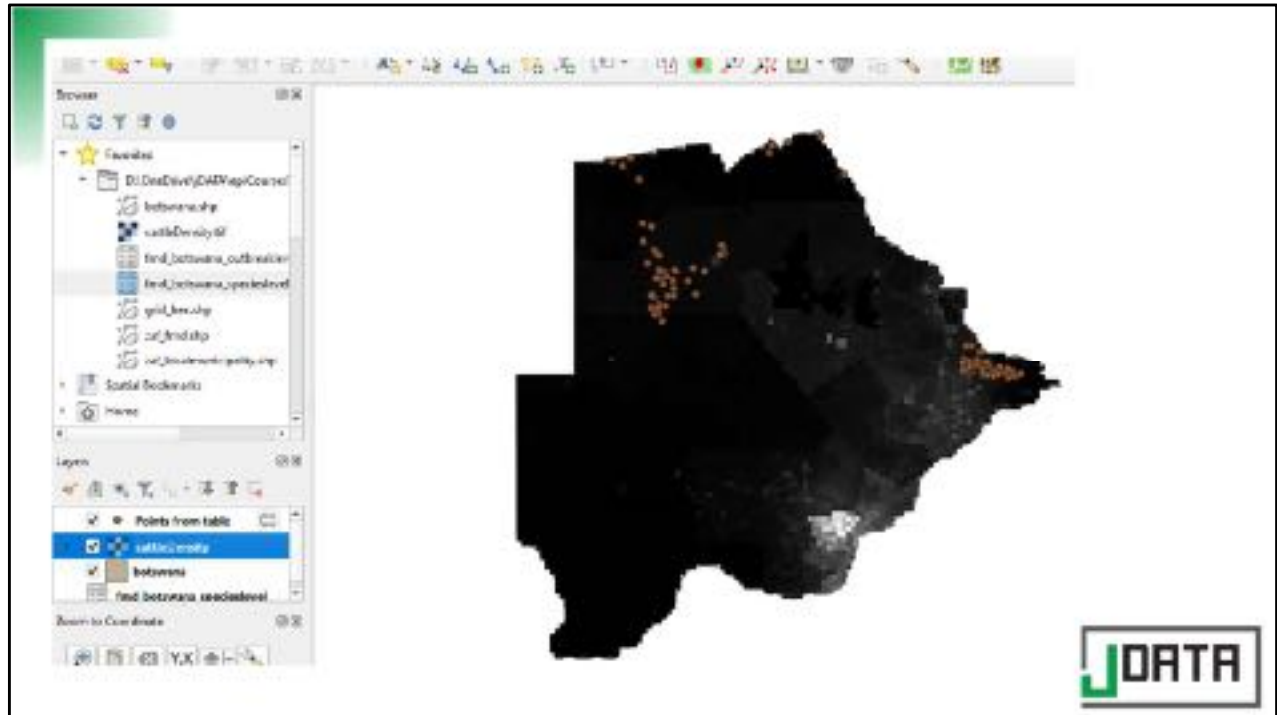


To see the point data above the polygon data click and drag the point data above the layer as the order is how they are rendered



- Click LAYER > Add Layer > Add Raster Layer

- 48

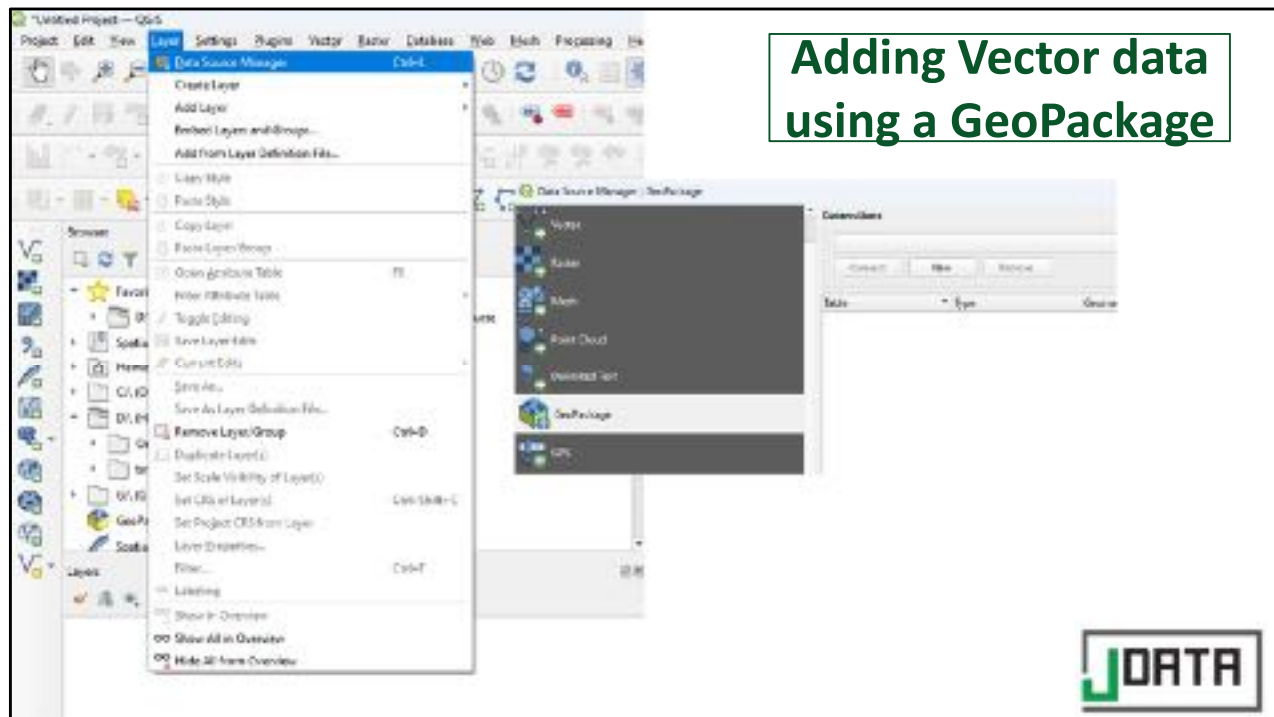


Ensure now that your Raster layer is sitting in between your Point (ABOVE) and Polygon (BELOW)

Adding Vector data using a GeoPackage

- The GeoPackage open format is a container that allows you to store GIS data (layers) in a single file.
- The styling of each layer can also be retained





- A connection to the GeoPackage needs to be made
- Layer > Data Source Manager
- Select GeoPackage
- Select NEW
- Browse to the Geopackage file

Adding Vector data using a GeoPackage

The screenshot illustrates the process of adding vector data using a GeoPackage in QGIS. On the left, the 'CONNECTIONS' panel shows a table of available connections. On the right, the 'BROWSER' panel shows the file system structure, including a 'GeoPackage' folder. An orange arrow indicates the flow of data from the connections to the browser.

Name	Type	Geometry column
between	MultiPolygon	geom
grid_hex	MultiPolygon	geom
zsf_hex	Point	geom
zsf_municipality	MultiPolygon	geom

- Once the connection is made a list of layers is shown
- Close the Connections and in the BROWSER scroll to available GeoPackage connections and now drag and drop the layers into the LAYERS section
- The styling that was saved with that Package is now also the default styling of the layer, although it can still be changed

TASK 3.1 – 30 min

- Basic qGIS operations
 - Open qGIS and familiarise yourself with the program
 - Import the *fmd_botswana_outbreakLevel.csv* file from the *data\gis\inCourse* folder and depict the locations as point vectors
 - Import the *botswana* country boundary polygon from the same folder
 - Import the *cattleDensity.tif* raster from the same folder
 - Connect to the *incourse.gpkg* GeoPackage and import the *grid_hex* vector from that GeoPackage
 - Make a screenshot of your layers window (WINKEY+SHIFT+S) or PrintScreen, open it in a picture editor like PAINT, and save as an image. Upload this to EpiCollect5 under **Task 3.1**



EpiCollect

- <https://five.epicollect.net/> and login and register if you haven't already – GMAIL is easier
- <https://five.epicollect.net/project/gis-assessments/data> and click on ADD ASSESSMENT



- Click RIGHT through any assessments that are not applicable, you can edit the form later for when these are done
- Save and Exit

